

SHIELD: Failure-Aware Shielding for Vision-Language-Action Models

Daniel Contreras-Esquivel Tianhui (Alina) Huang Jacob Lee

Department of Computer Science, Stanford University

Problem, Prior Work & Architecture

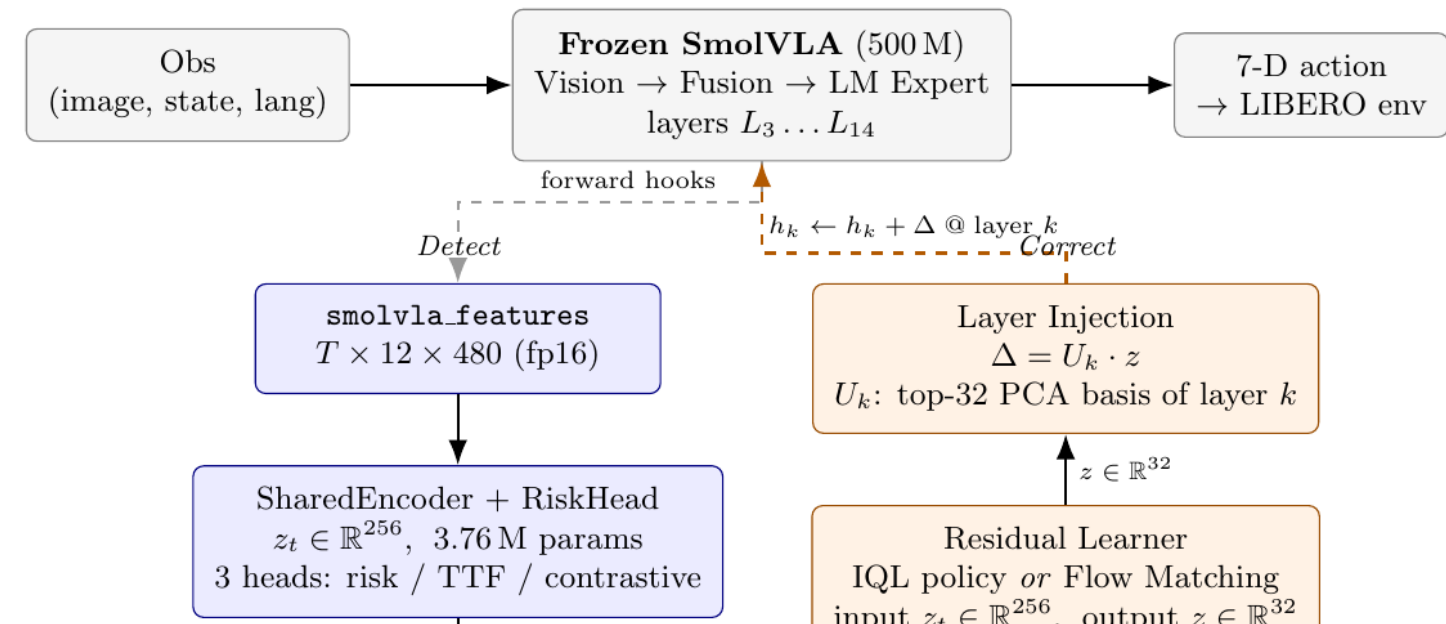
Problem. VLA policies fail *silently* under distribution shift — sensor noise, viewpoint drift, and action jitter corrupt hidden states with no detect-or-correct loop.

SHIELD is the first *triggered, learned*, activation-space shield for VLAs: fires only when calibrated risk exceeds a threshold; injects a flow-model correction; falls back safely when inactive. No base weights modified.

Research question. Can we detect impending manipulation failures from VLA activations and inject a lightweight residual to recover success, without retraining the base model?

Prior work

- **Activation steering** (RepE, ITI, ROME) — steers LLMs via residuals. None address continuous-action VLAs.
- **VLA robustness** (VLA-OS, Cerebellar, ReSteer) — layer analysis & action-space correction. Not triggered.
- **COAST** (concurrent) — always-on conceptor steering ($h' = h \cdot M$). SHIELD adds calibrated triggering and a learned flow model.



Data & Perturbations

5,117 labeled rollouts of SmolVLA on LIBERO across 11 perturbation types. Label: fail_in_H ($H = 30$) + per-episode failure-type tag.

Split	Contents
Eval only	Joint-angle noise (hard negatives)
Train + eval	Action noise, viewpoint shift, Gaussian blur + windowed variants ($\times 2$ each)
Trigger retrain	999 windowed + 250 nominal successes (timeout failures excluded)
Held out	4,118 continuous perturbations

Labels. Binary fail_in_H ($H = 30$) + coarse failure-type tag (collision, drop, grasp misalignment, grasp precision) — the NT-Xent supervision signal for z_t geometry. Windowed perturbations give clean risk-on/off transitions per episode; nominal timeout failures excluded from retraining.

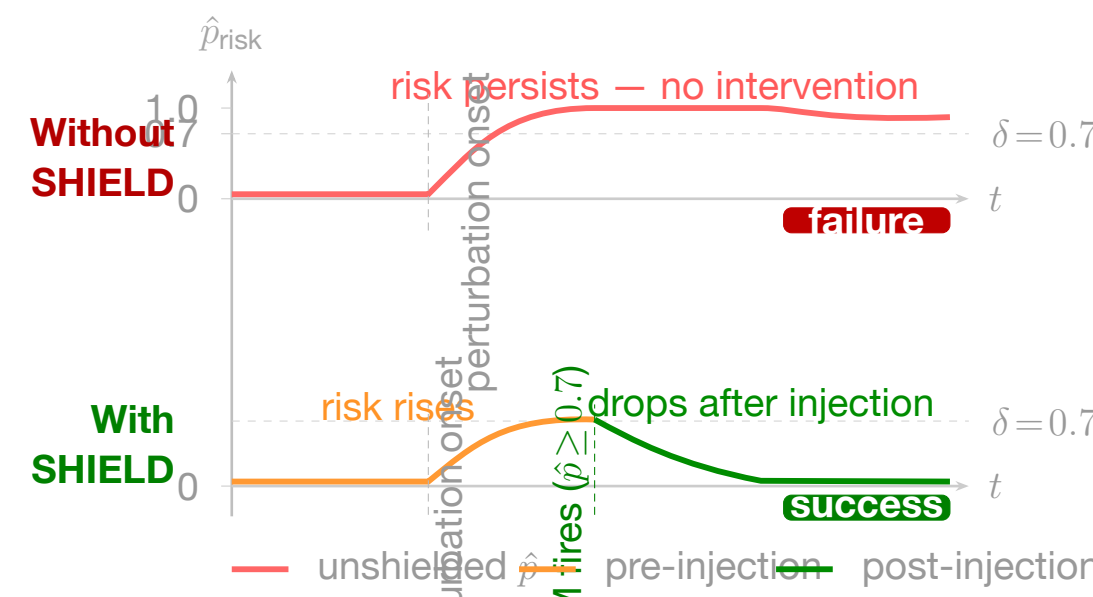
Research Paradigm

Hypothesis. Failure is encoded in VLA activations *before* it manifests in actions. A calibrated trigger plus a learned flow residual can redirect the policy back to success without modifying the base model.

Question	Operationalization
VLA policies fail silently. Can we read impending failure from VLA activations and	Step 1 — Tap & detect. Hooks at layers $\{4, 5, 6, 11, 12, 13\}$; SharedEncoder $\rightarrow \hat{p}_{\text{risk}}, z_t \in \mathbb{R}^{256}$. Step 2 — Failure-type geometry. BCE + NT-Xent clusters z_t by failure type for flow conditioning. Step 3 — Gate, inject, measure. FSM: enter $\hat{p} \geq 0.7$, exit $K = 5$ steps below 0.4. $\Delta = U_k z, h'_k = h_k + \Delta$.

Success criterion. Shielded SR > unshielded SR on perturbed episodes; clean SR must not degrade. Post-retrain: action noise +5 pp (0.75 $\rightarrow$ 0.80); high FPR ($\approx 0.75 - 0.79$) drives over-injection on clean episodes.

SHIELD in Action ($\hat{p}_{\text{risk}}$ vs. time):



Risk Head Architecture

SharedEncoder (3.76M params): temporal CNN over 6 hooked layers + cross-layer attention $\rightarrow$ two outputs:

- $\hat{p}_{\text{risk}} \in [0, 1]$ — calibrated failure prob
- $z_t \in \mathbb{R}^{256}$ — failure-type latent for flow conditioning

$$\mathcal{L} = \lambda_{\text{BCE}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{TTF}} \mathcal{L}_{\text{TTF}} + \lambda_{\text{NT-Xent}} \mathcal{L}_{\text{NT-Xent}}$$

FSM & injection. Enter: $\hat{p} \geq 0.7$; exit: $K = 5$ steps below 0.4. Flow model: $\Delta = U_k z, h'_k = h_k + \Delta$ via top-48 PCA basis. No VLA weights modified.

Risk Head Ablation

Run	wAUC	Silh.	ECE	BSkill
A BCE only	0.928	-0.046	0.088	-0.067
B BCE+TTF	0.927	-0.040	0.093	-0.12
C BCE+TTF+NT-Xent	0.917	+0.168	0.077	+0.018
D NT-Xent only	0.440	+0.125	—	—

Table 1. Loss ablation (Run C selected, highlighted).

NT-Xent costs -0.010 wAUC vs. B while raising silhouette $-0.040 \rightarrow +0.168$ and improving calibration (ECE 0.093 $\rightarrow$ 0.077). Better geometry enables failure-type conditioning in the flow model — a deliberate design trade-off.

val z_t — UMAP (silhouette=0.168)

Layer Probe & PCA Diagnostics

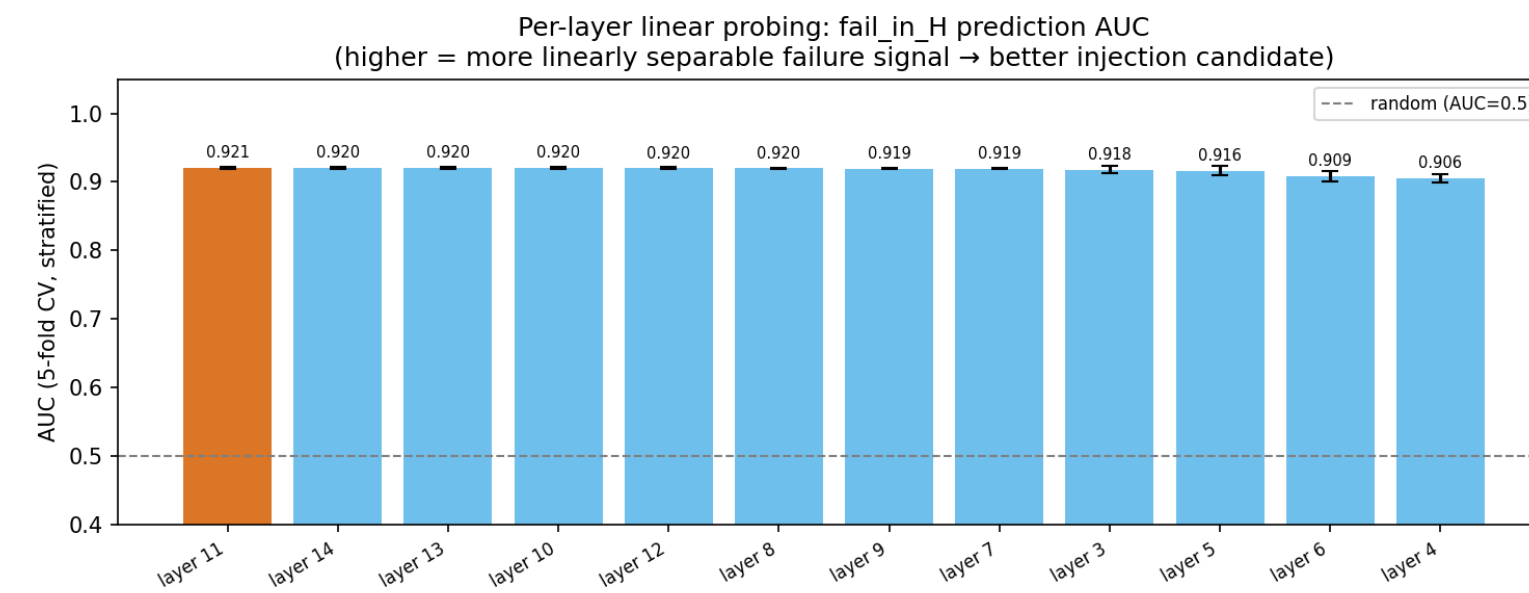


Figure 3. Linear probe AUC per action-expert layer (≈ 0.92 flat across all 12). Failure is linearly separable *everywhere* in SmolVLA's activations — the signal is pervasive, not layer-specific. Layer 11 selected: mid-depth lets downstream layers regenerate coherent actions from the corrected state.

Shield Evaluation

Post-retrain eval: 80 closed-loop LIBERO episodes (2 tasks $\times$ 10 ep $\times$ 4 conditions), retrained Run C risk head ($T = 1.34$), layer-11 injection.

Perturbation	Unsh.	SHIELD	Δ SR	Prec.	FPR
none	0.80	0.60	-0.20	—	—
action noise	0.75	0.80	+0.05	0.258	0.787
image blur	0.70	0.65	-0.05	0.233	0.772
joint noise	0.85	0.75	-0.10	0.320	0.752

Table 2. Post-retrain shield eval (2 LIBERO-Spatial tasks, $n=20/\text{cond}$). Action noise is the only win; high FPR drives over-injection on clean episodes.

Key diagnostic. Action noise: SHIELD raises SR 0.75 $\rightarrow$ 0.80 (+5 pp). False-positive injections on clean episodes dominate failure cases (FPR $\approx 0.75 - 0.79$ across perturbation types).

Insights & Takeaways

- **Failure is accessible in VLA activations.** Linear probe AUC ≈ 0.92 across *all* 12 action-expert layers — signal is pervasive and available from the first inference step.
- **Contrastive geometry is the key design choice.** NT-Xent raises silhouette $-0.040 \rightarrow +0.168$ at only -0.010 wAUC cost, giving the flow model typed failure conditioning rather than a scalar risk score.
- **Residual injection recovers success on action noise.** Post-retrain SHIELD: action noise +5 pp (0.75 $\rightarrow$ 0.80 SR) across 20 episodes; other perturbations show neutral or negative Δ SR.
- **Bottleneck: trigger precision.** FPR $\approx 0.75 - 0.79$ causes over-injection on clean and perturbed episodes. Tighter threshold or cooldown steps is the clearest next lever.

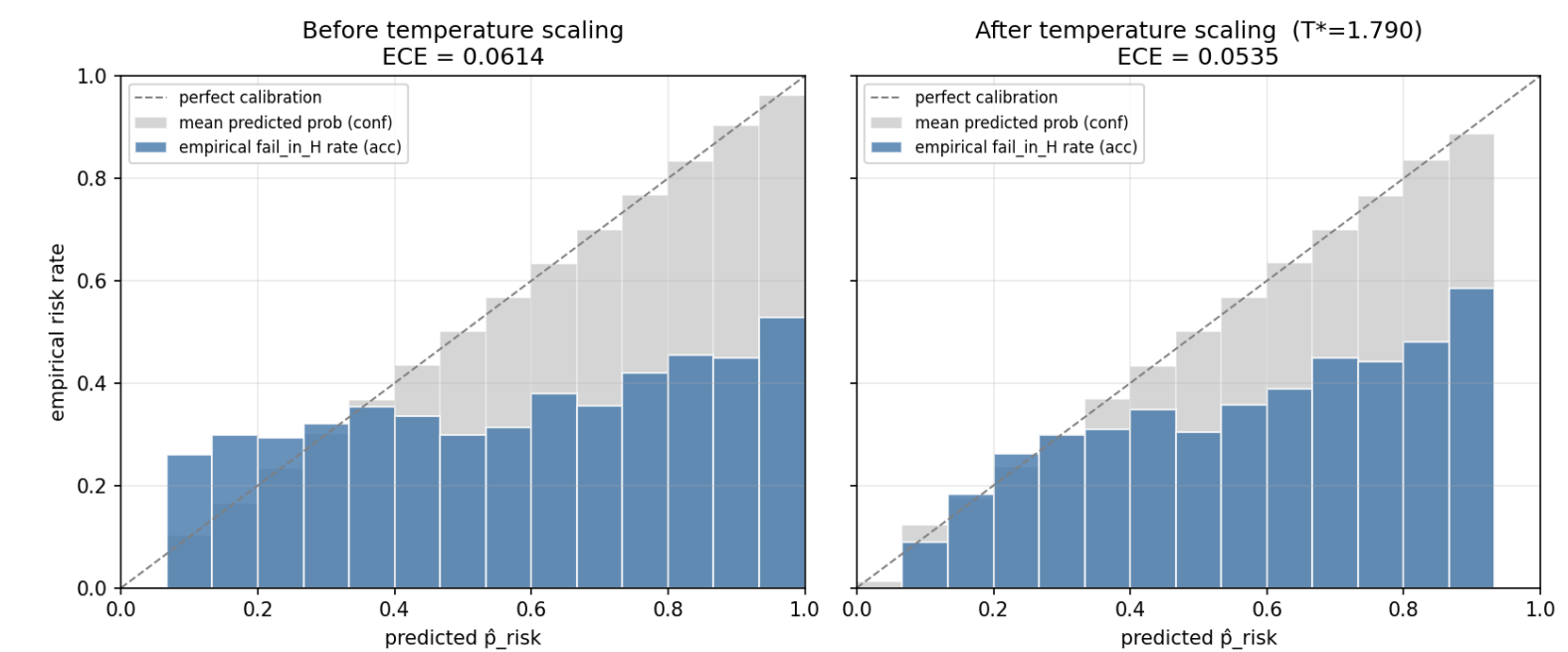


Figure 4. Run C (retrained) reliability diagram after temperature calibration ($T=1.34$). ECE 0.040; Brier 0.053. Well-calibrated $\hat{p}_{\text{risk}}$ makes the hysteresis threshold meaningful.

Limitations, Future Work & Code

Limitations

- Simulation only (LIBERO); real-world not validated
- Single VLA (SmolVLA 500M)
- Eval on 2/4 LIBERO tasks; tasks 1 & 6 in progress
- U_k vs. U_{null} vs. U_{potent} not yet compared

Future work

- Failure-type routing & multi-task VLAs
- Real-world transfer
- Online threshold adaptation



github.com/TinaLand

?tab=repositories

Risk head code, activation hooks, flow model (C12/C13), rollout videos, figures & sweep plots.